Architecture.md v1.1

Chapter 10 — Logging, Monitoring & Observability

This chapter defines how AI OS records system activity, monitors component health, and provides observability for debugging, auditing, and performance analysis.

Logging is a **Core Service**. Every module in AI OS must produce standardized logs.

10.1 Purpose

The Logging & Monitoring System shall:

- Record all significant system activity.
- Track execution performance.
- Detect failures.
- Assist debugging.
- Provide audit information.
- Monitor system health.
- Support future analytics dashboards.

Logging is observational only. It never performs business logic.

10.2 Design Philosophy

The system follows three principles:

Complete

Every important operation is logged.

Structured

All logs follow one common schema.

Lightweight

Logging should not noticeably reduce system performance.

10.3 Log Categories

The system supports the following categories.

System Logs

Examples:

- Startup
- Shutdown
- Configuration changes

Request Logs

Examples:

- User request received
- Request completed
- Processing duration

Agent Logs

Examples:

- Agent initialized
- Agent execution
- Agent failure
- Agent health status

Memory Logs

Examples:

- Memory read
- Memory update
- Memory archive

Security Logs

Examples:

- Permission granted
- Permission denied
- Approval requested
- Approval completed

Database Logs

Examples:

- Query execution
- Transaction commit
- Rollback
- Connection failure

Workflow Logs

Examples:

- Workflow started
- Workflow paused
- Workflow completed
- Workflow cancelled

10.4 Standard Log Format

Every log entry shall include:

- Log ID
- Timestamp
- Request ID
- Component Name
- Event Type
- Severity
- Status
- Execution Time
- Message

Optional:

- Exception Details
- Stack Trace (Debug Mode Only)

10.5 Severity Levels

Four severity levels are defined.

INFO

Normal system operations.

Example:

Agent Started

WARNING

Unexpected but recoverable situations.

Example:

Retry initiated.

ERROR

Operation failed.

Example:

Database unavailable.

CRITICAL

System integrity is at risk.

Example:

Memory corruption.

10.6 Monitoring Service

The Monitoring Service continuously tracks system health.

Metrics include:

- CPU Usage
- Memory Usage
- Disk Usage
- Database Health
- API Latency
- Active Workflows
- Active Agents
- Queue Length

The Monitoring Service does not control system execution.

10.7 Health Checks

Every major component implements a health check.

Components include:

- Router
- Jarvis
- Friday
- Agent Manager
- Memory Service
- Worker Agents
- Database
- Event Service

Health states:

- HEALTHY
- DEGRADED
- OFFLINE
- FAILED

10.8 Performance Metrics

The following metrics are collected.

Per Request:

- Total Duration
- Agent Count
- Memory Reads
- Memory Writes
- Approval Wait Time

Per Agent:

- Average Execution Time
- Failure Rate
- Success Rate
- Retry Count

Per System:

- Requests Per Minute
- Active Sessions
- Uptime

10.9 Error Reporting

Errors shall include:

- Request ID
- Agent
- Error Code
- Description
- Timestamp

Internal implementation details are hidden from end users unless Debug Mode is enabled.

10.10 Log Retention

Logs are categorized by retention policy.

Debug Logs

Short-term retention.

Automatically deleted after the configured period.

Operational Logs

Retained for diagnostics.

Archived after expiration.

Audit Logs

Long-term retention.

Cannot be modified.

Deletion requires administrative approval.

10.11 Dashboard Metrics

Future dashboard displays may include:

- System Status
- Active Agents
- Request Timeline
- Agent Performance
- Memory Usage
- Workflow Queue
- Security Alerts
- Error Trends

Dashboard reads monitoring data only.

10.12 Alerting

The Monitoring Service may generate alerts.

Examples:

- Database offline
- Agent repeatedly failing
- High memory usage
- Permission abuse
- Workflow timeout

Alerts are classified as:

- Low
- Medium
- High
- Critical

10.13 Debug Mode

Debug Mode is intended for development only.

Additional information may include:

- Stack traces
- Internal payloads
- Execution path
- Timing breakdown

Debug Mode must be disabled in production by default.

10.14 Privacy Rules

Logs must never expose:

- Passwords
- API Keys
- Authentication Tokens
- Encryption Keys
- Sensitive personal data

Sensitive values should be masked or omitted.

10.15 Monitoring Rules

The Monitoring Service may:

- Observe
- Record
- Analyze
- Notify

It must never:

- Execute business logic
- Modify memory
- Route requests
- Invoke worker agents

10.16 Future Expansion

Future versions may support:

- Real-time monitoring dashboard
- Distributed tracing
- Metrics export
- Predictive failure detection
- AI-assisted diagnostics
- Performance benchmarking
- Anomaly detection

These additions must remain compatible with the standardized logging architecture.

10.17 Design Rules

The Logging & Monitoring System must satisfy:

- Complete observability.
- Standardized log format.
- Centralized monitoring.
- Immutable audit records.
- Privacy protection.
- Low performance overhead.
- Independent operation.

No component may implement its own incompatible logging system.

End of Chapter 10

Next Chapter: Chapter 11 — Configuration Management & Environment Architecture